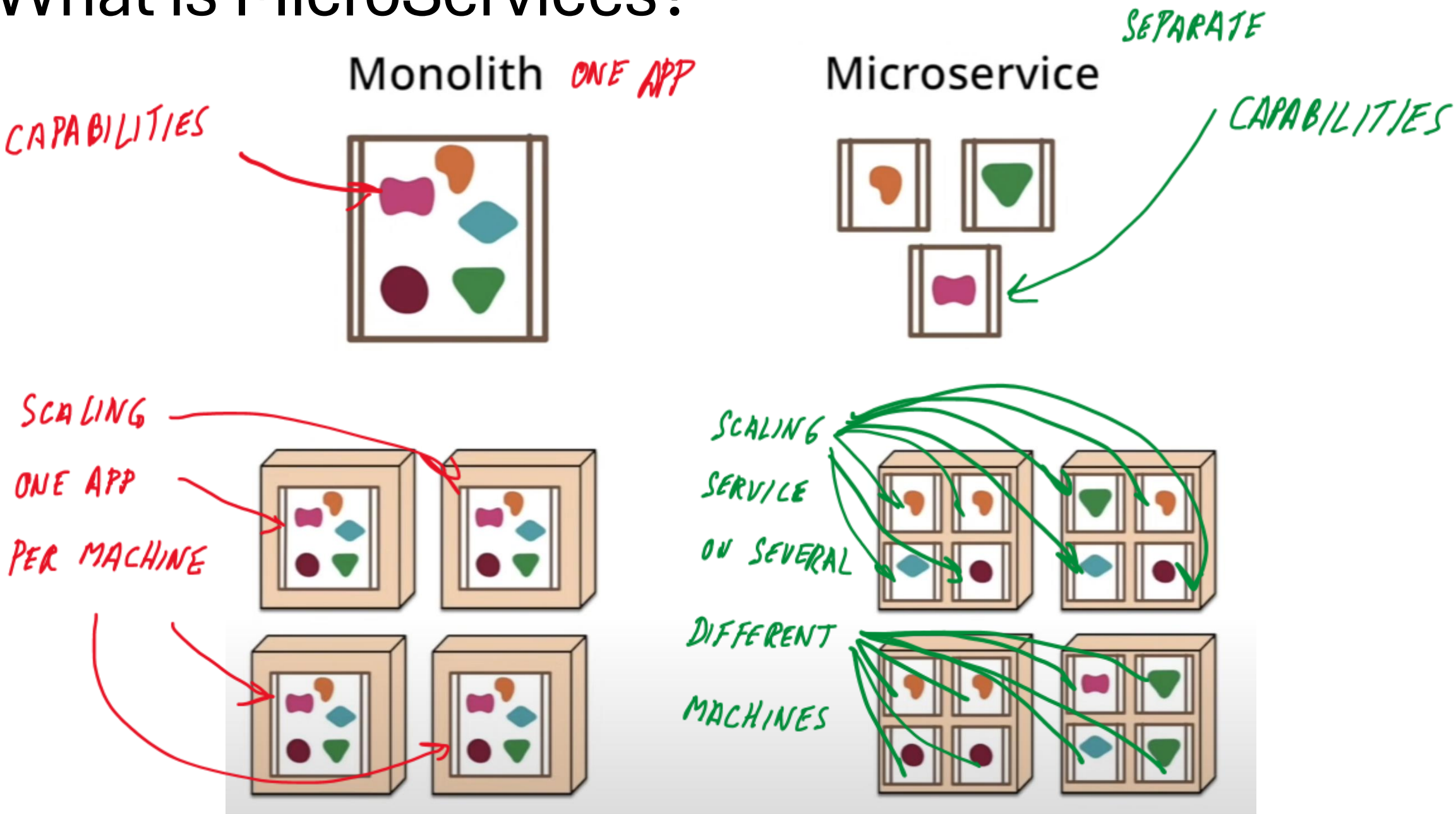
MicroServices

Jan C. Sørensen

What is MicroServices?



MicroServices Characteristics

1. Componentization via services
2. Organized around business capabilities
3. Products not Projects
4. Smart endpoints and dumb pipes
5. Decentralized Governance
6. Decentralized Data Management
7. Infrastructure Automation
8. Design for failure
9. Evolutionary Design

Componentization via Services

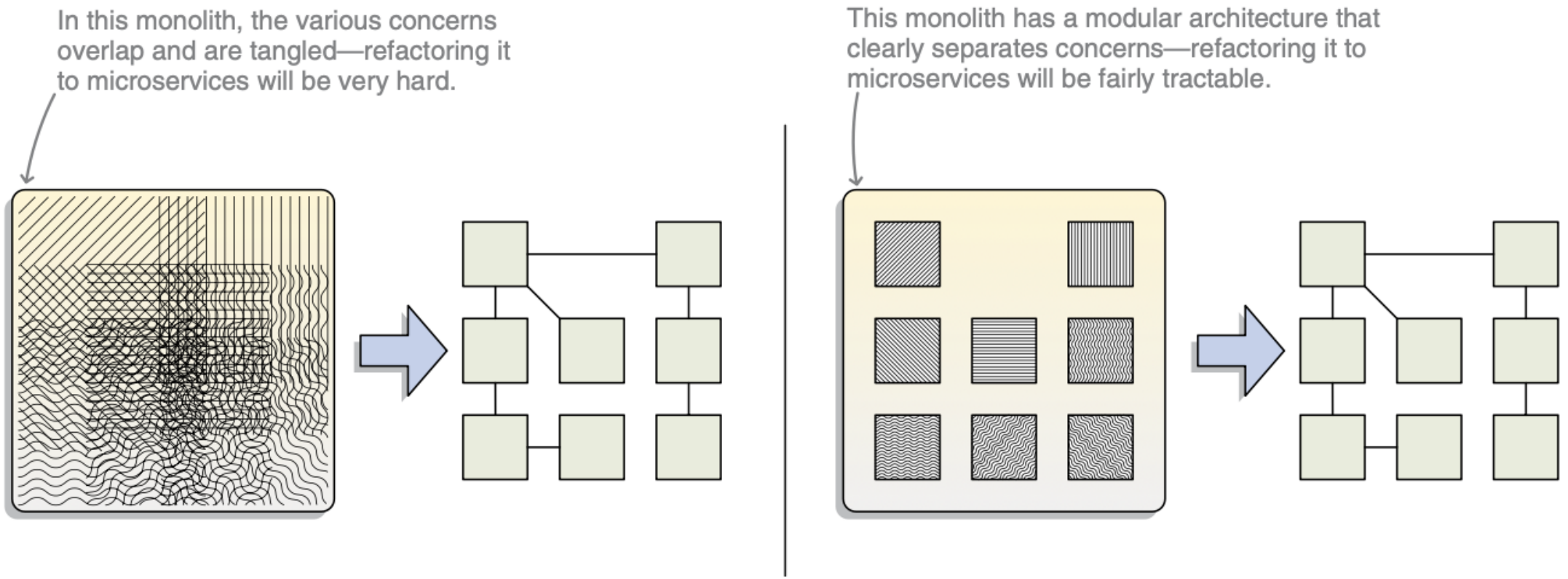
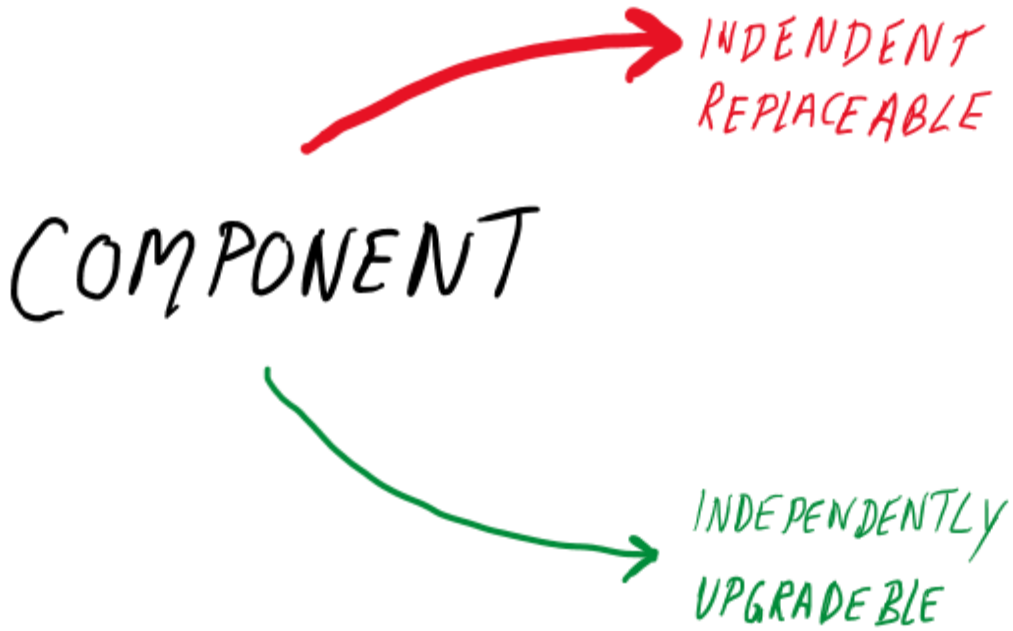
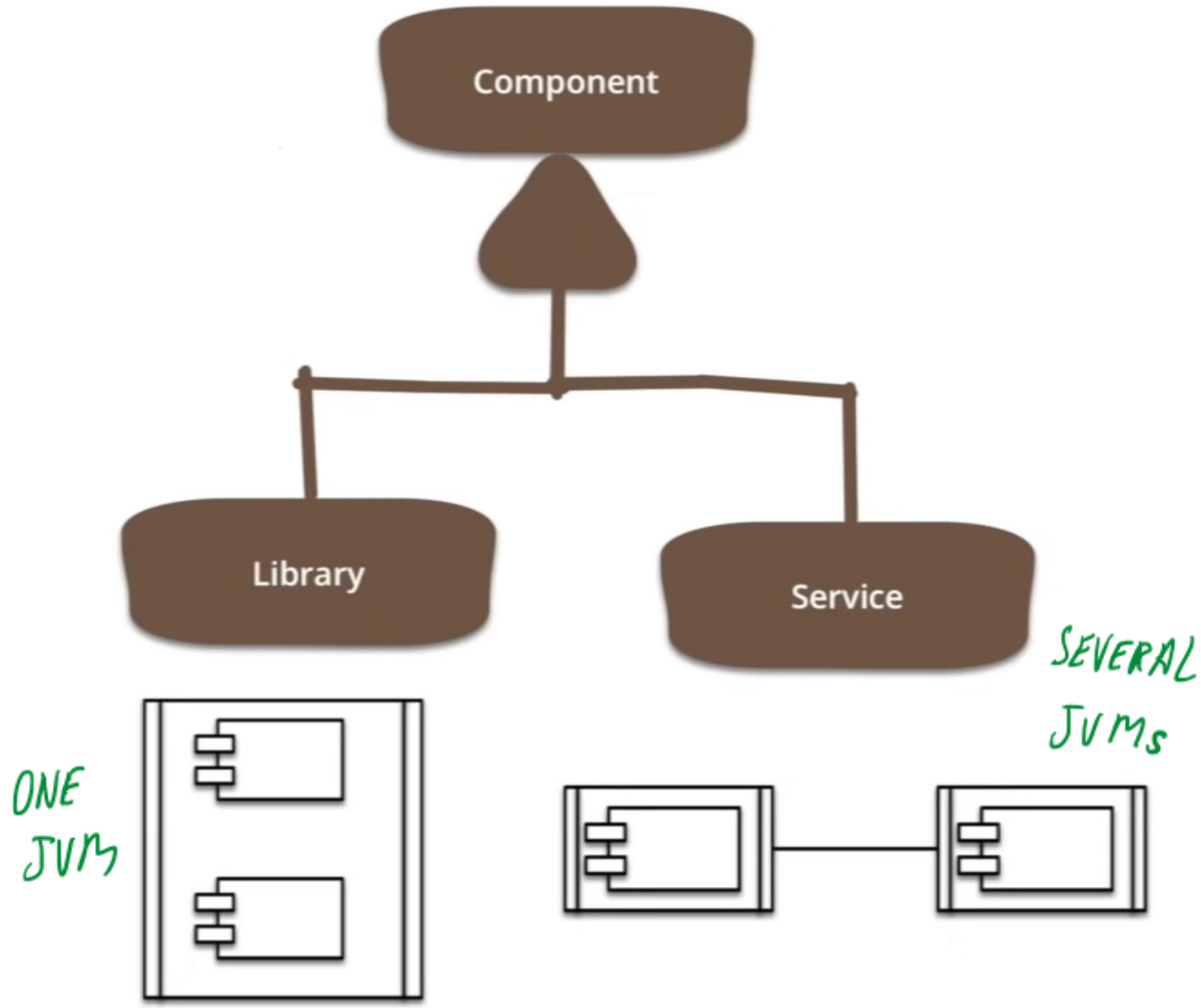


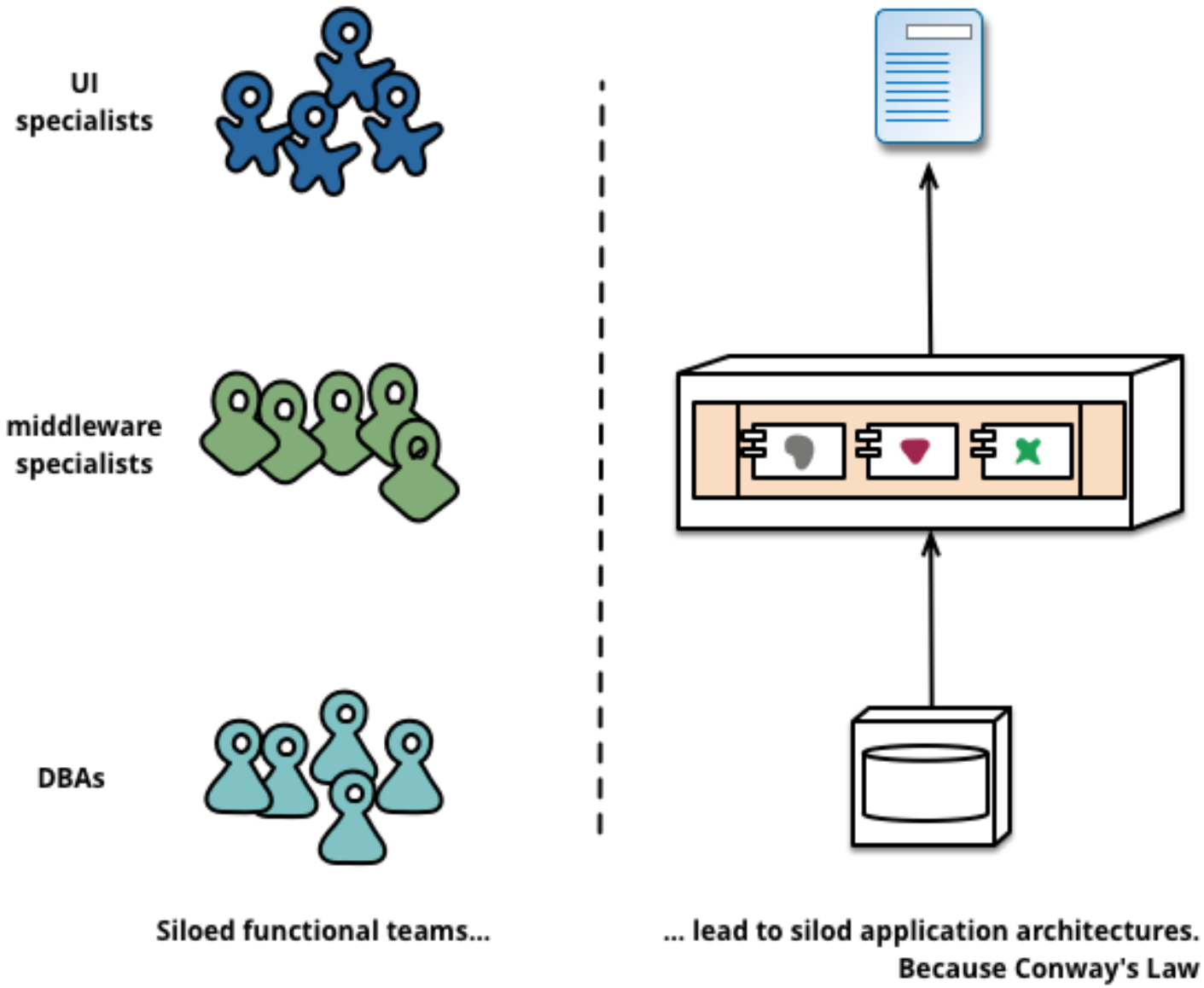
Figure 15.6 Given two hypothetical migrations of a monolithic application to microservices, would you rather start with a sizable square of mud (left) or a properly modularized code base (right)?

Componentization via Services

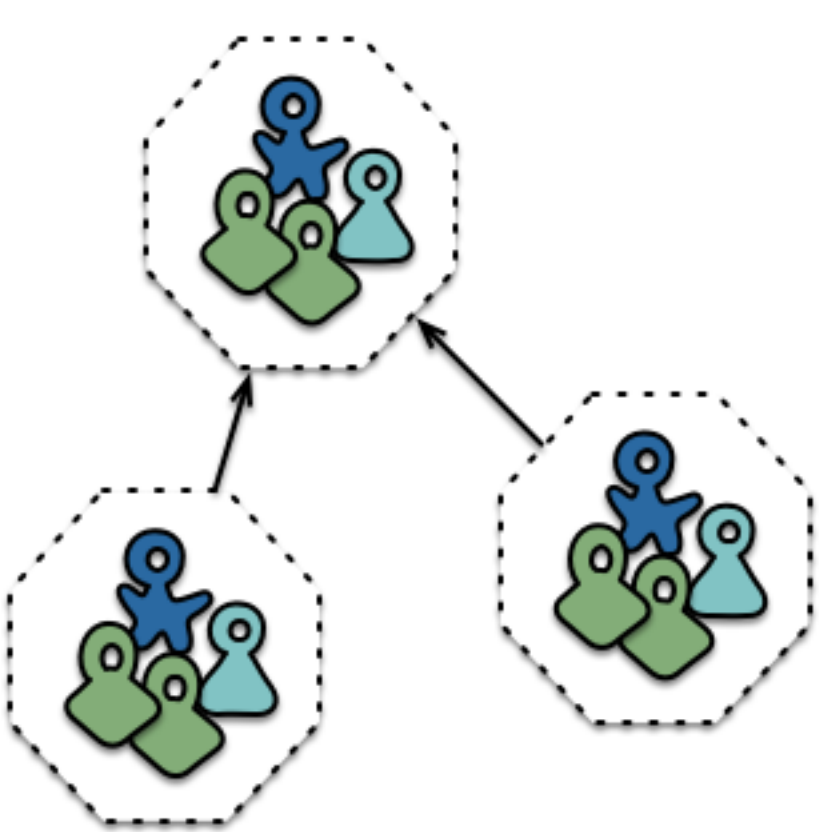




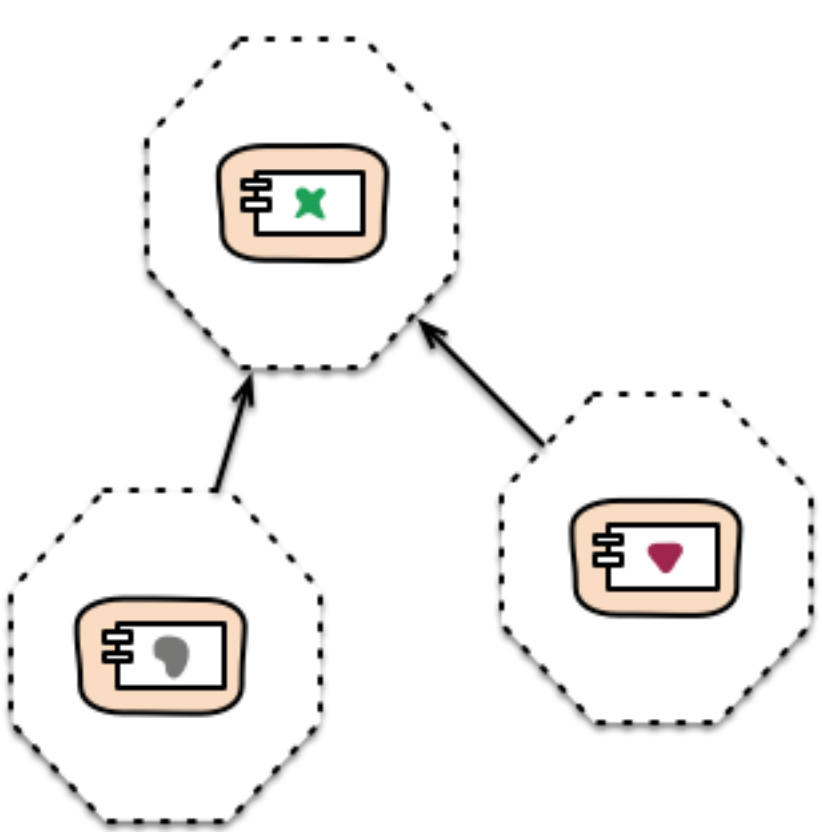
Organization of Monolith



Organization of MicroServices & CBSE



Cross-functional teams...



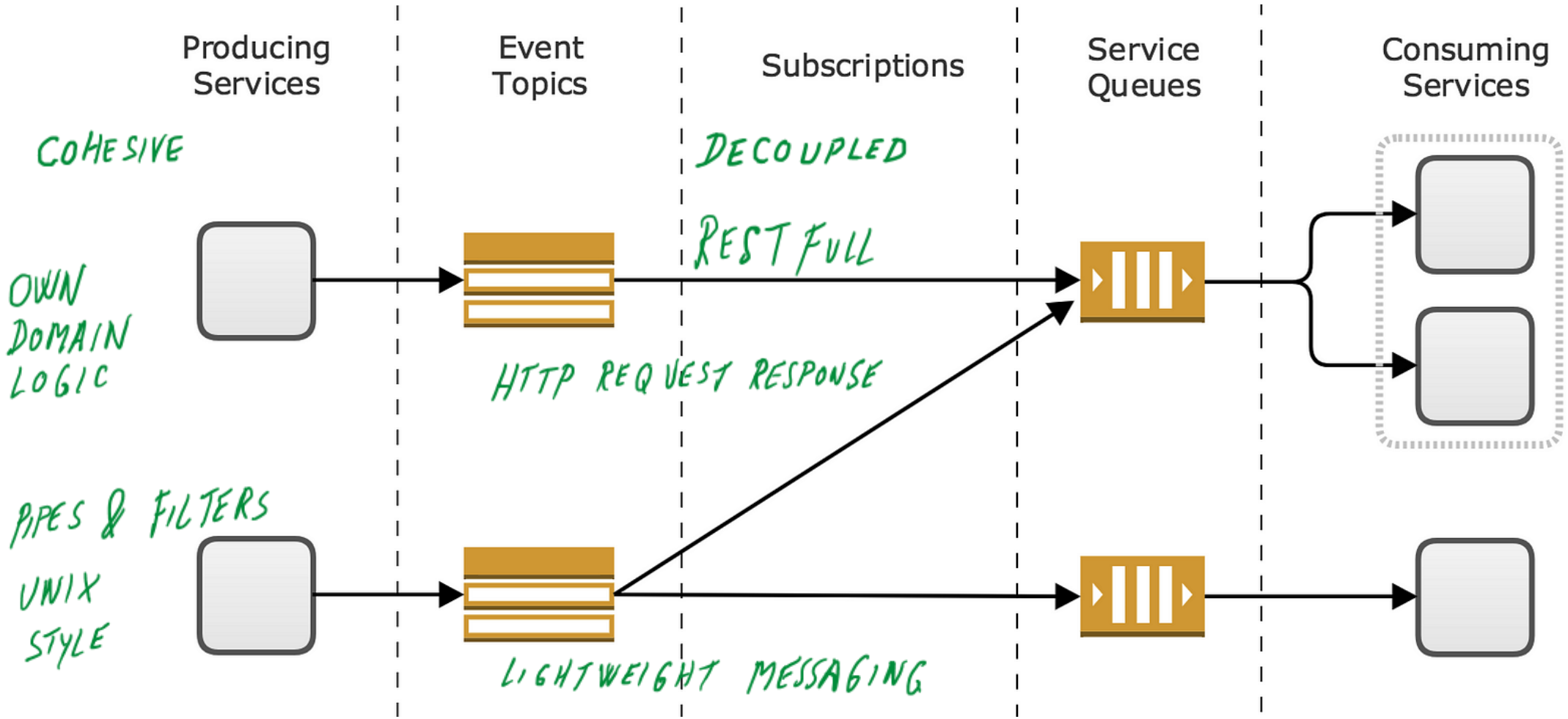
... organised around capabilities

Because Conway's Law

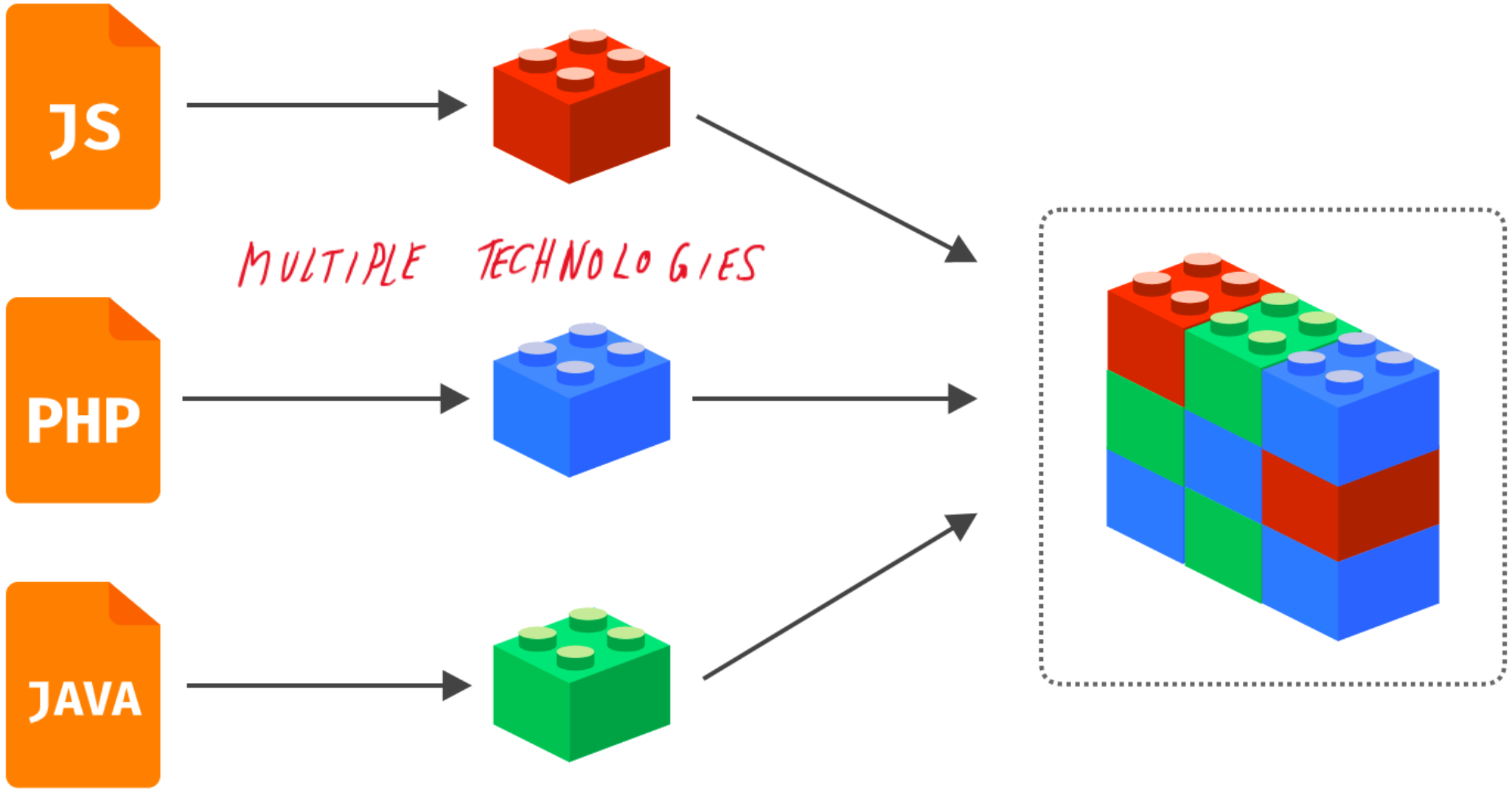
Product not Projects ("you build, you run it")

Product	**Project**
A way of answering a specific need	A way of executing a plan
Long term	Short term
Focus on creating a sustainable environment	Focus on fixed constraints (scope, resources, time, budget)
No fixed end date, continuous discovery and development needed	Has a defined start and end date
Unpredictable	Appears predictable (although this is often false)
Team stays together long term with slight changes over time	Team stays for the duration of the project

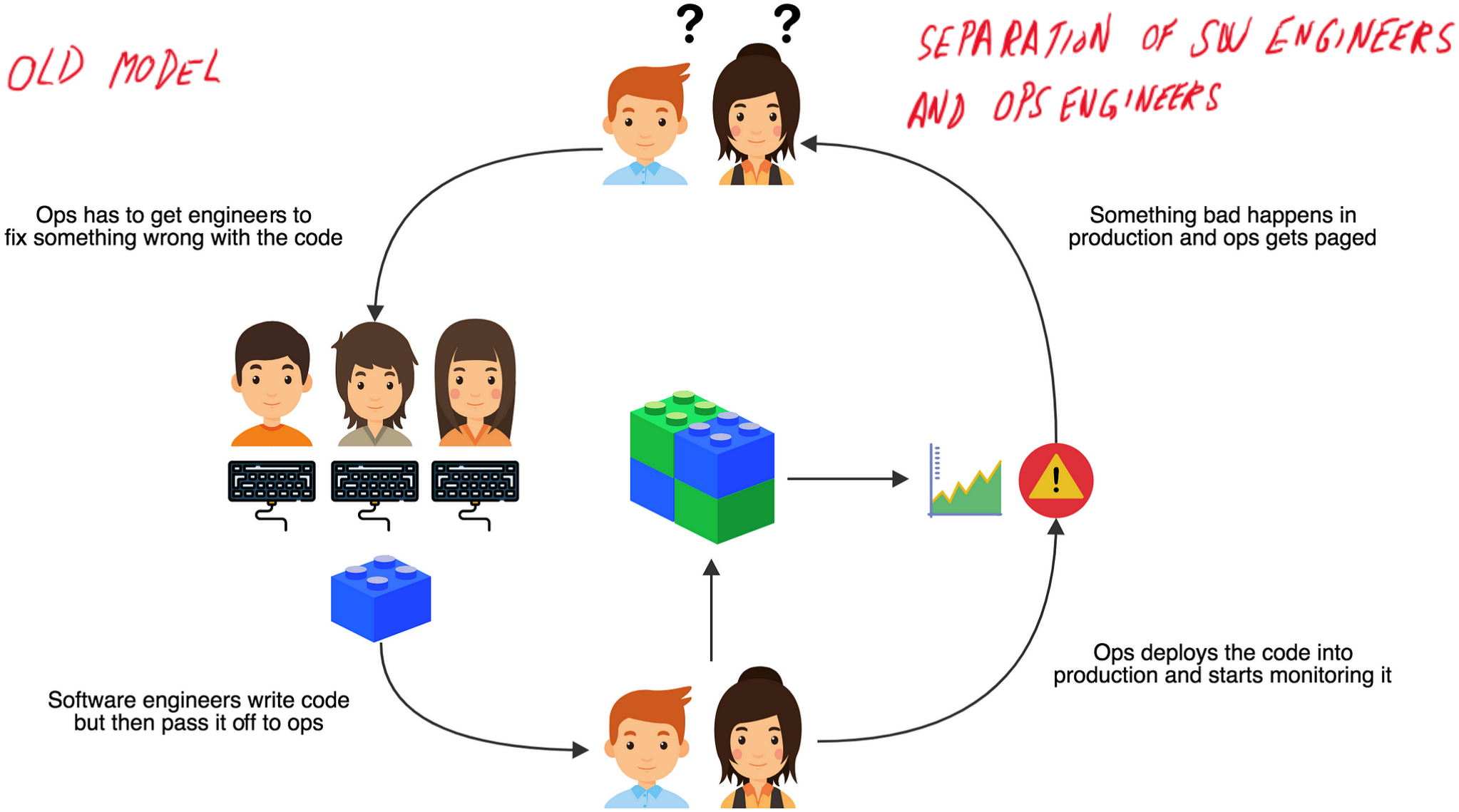
Smart Endpoints and Dumb Pipes



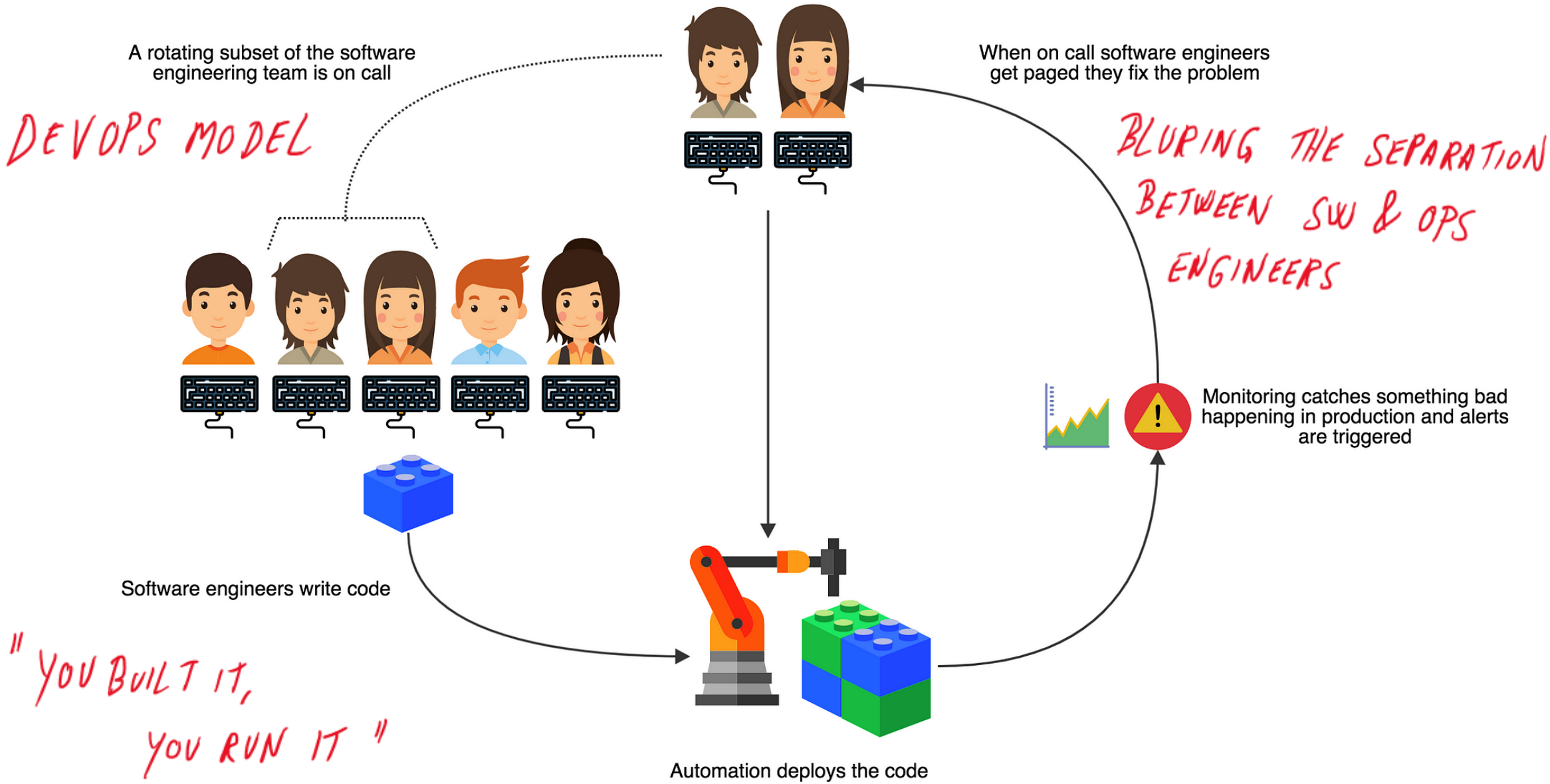
Decentralized Governance (Tech)



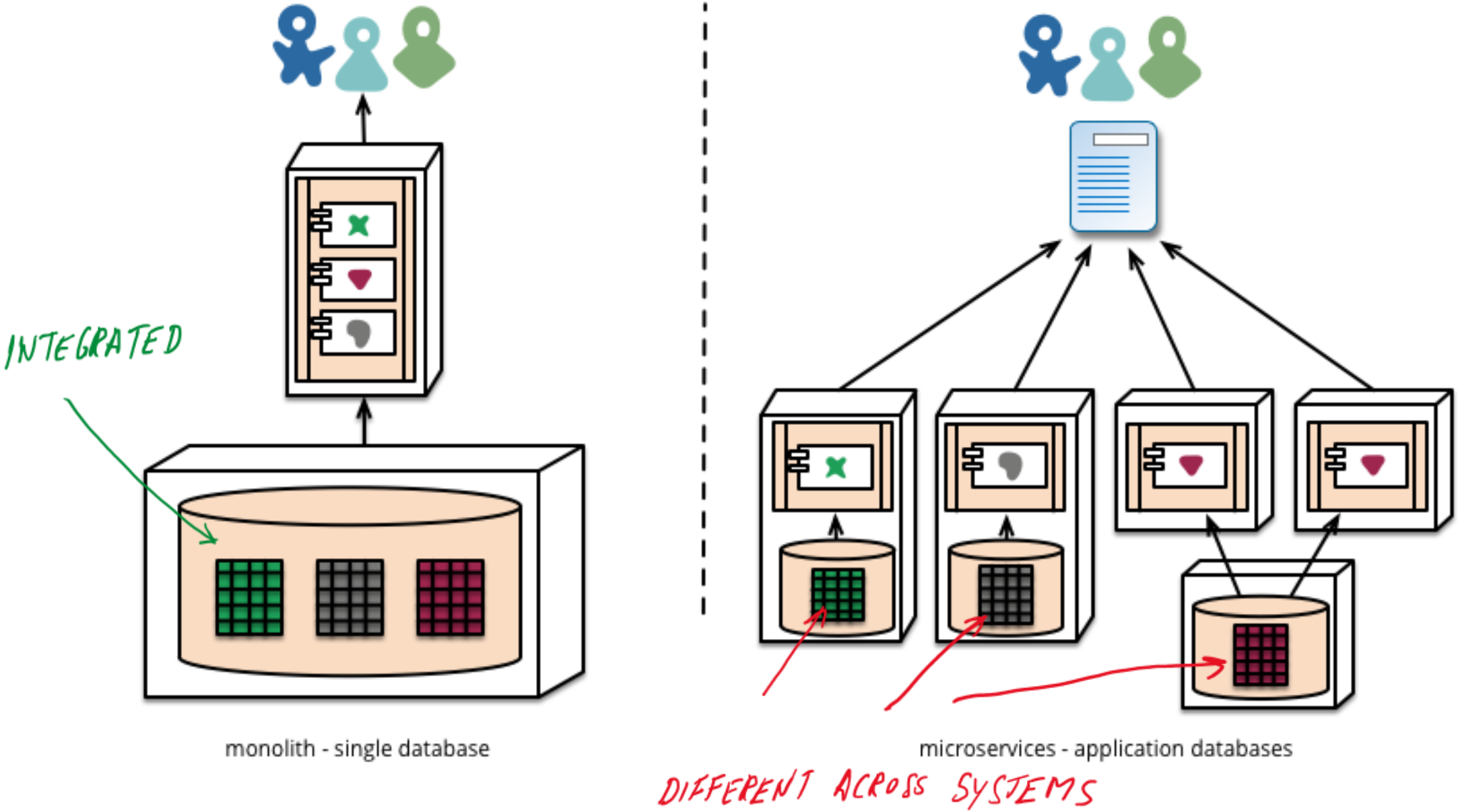
Centralized Governance (OPS)



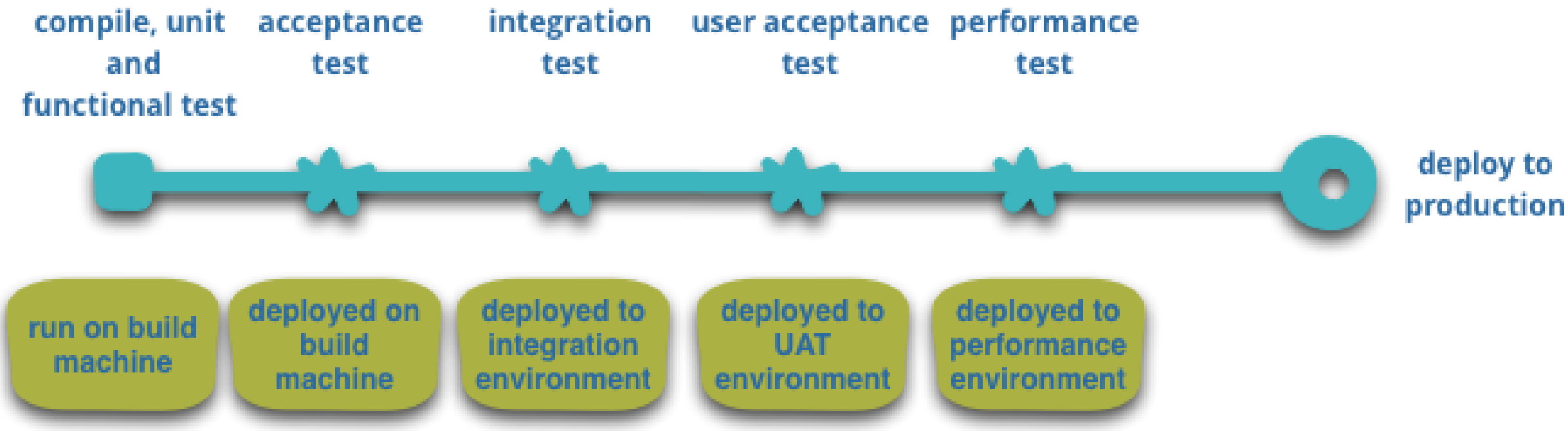
Decentralized Governance (DevOps)



Decentralized Data Management



Infrastructure Automation



Design for Failure

SERVICES CAN FAIL AT ANY TIME

↓

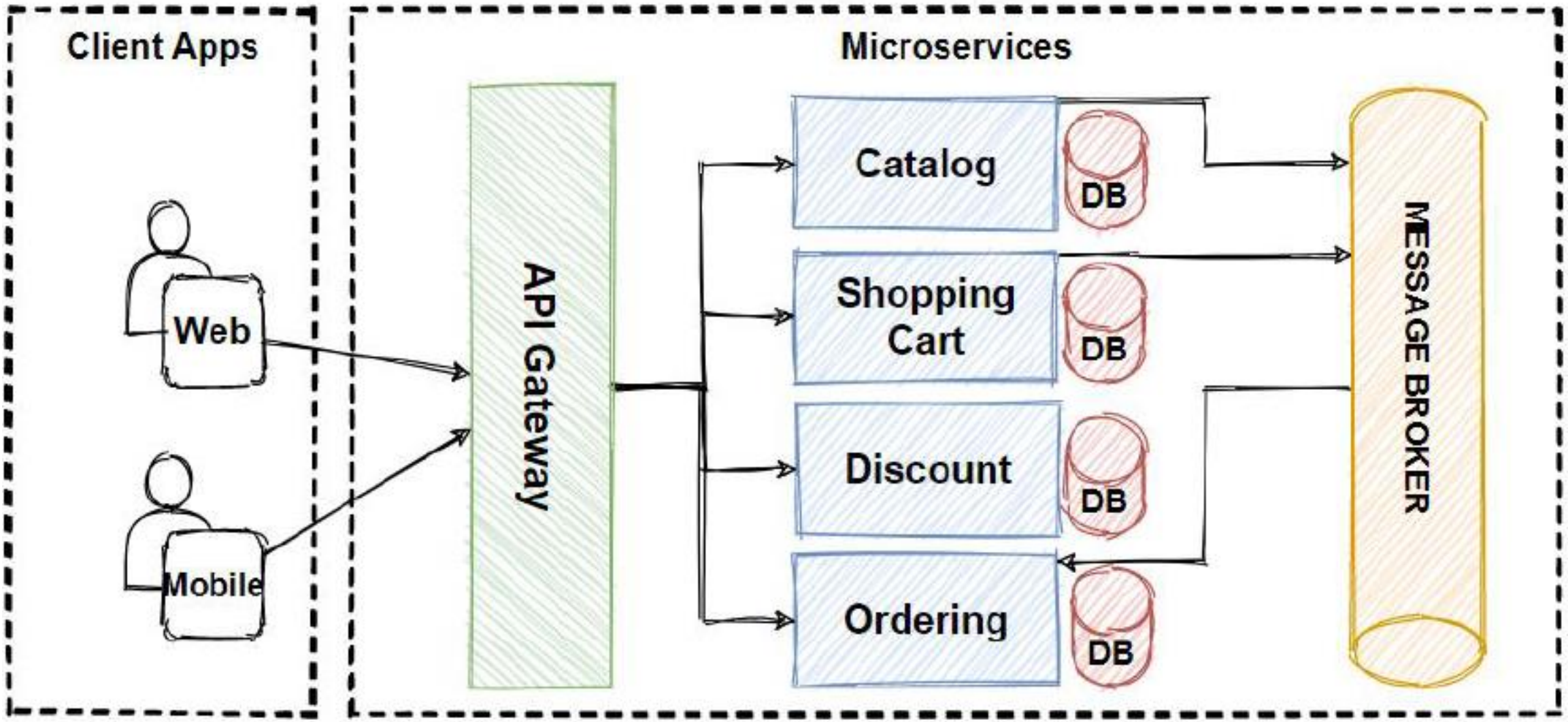
DETECT FAILURES QUICKLY

↓

SOPHISTICATED MONITORING



Evolutionary Design



Conclusion


You must be this tall to use microservices
Traceable Business Transactions
Continuous Delivery
Product-centered Teams
Multi-Dev Environment
Rapid Provisioning
Basic Monitoring
Rapid Application Deployment
Devops Culture

References

- Microservices by Martin Fowler
  https://martinfowler.com/articles/microservices.html

- The Java Module System by Nicolai Parlog
  https://www.manning.com/books/the-java-module-system